

Alex Liu

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Education

University of Florida (UF) - Gainesville, FL

Expected Graduation: **May 2028**

Bachelor of Science in Computer Science, GIS Certificate

GPA: 3.75/4.0

Relevant Coursework: Data Structures & Algorithms, Computer Org, Discrete Structures, Intro to Software Engineering

Skills

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, Go, C#, Bash, HTML/CSS, RISC-V Assembly

Backend: FastAPI, Flask, Node.js, REST APIs, JWT, SQLAlchemy, Pydantic

Frontend: React, Tailwind CSS, Vite

Cloud/DevOps: AWS (Lambda, ECS, S3, RDS), Docker, Git, GitHub Actions (CI/CD), Linux/Unix, Postman

Data/ML: OpenCV, MediaPipe, NumPy, Pandas

Concepts: Microservices, Distributed Systems, API Design, OOP, Data Structures & Algorithms

Projects

Geography Dashboard, *React, AWS, Leaflet, FastAPI, Vite, TypeScript, PostgreSQL, Docker, Gemini AI* **Nov 2025**

- Built a **full-stack** geospatial analytics platform that ingests user-uploaded spatial datasets (CSV/GeoJSON) and generates DBSCAN clusters and density grids for city-scale insights.
- Designed a decoupled **microservices** architecture for **distributed deployment** across hybrid cloud (**AWS**), separating core GIS processing (**FastAPI + PostgreSQL**) from AI inference for scalability and fault isolation.
- Implemented end-to-end distributed tracing by propagating **X-Request-ID** across services, with structured logging for cross-service observability/monitoring in a hybrid-cloud environment.
- Automated containerized deployments with Docker & GitHub Actions (**CI/CD**), leveraging platform-managed deployments, and implemented rate limiting to control GenAI usage/cost and improve reliability under burst traffic.

UF Health SmartScribe, *HTML, CSS, JavaScript, Supabase, Gemini AI* **Oct 2025**

- Built an AI medical scribe prototype in a 24-hour designathon that converts clinician-patient audio into SOAP notes
- Implemented a real-time streaming pipeline using browser audio capture + **Deepgram** diarization, then a **Gemini** prompt chain to extract clinical facts and generate speaker-aware **SOAP drafts** (near real-time, ~5s prototype latency).
- Integrated **Supabase authentication** and **PostgreSQL** persistence for secure encounter storage/retrieval; reduced clinical documentation time by ~60% with human-in-the-loop editing.

SwampGuard — AI Insurance Claims Analysis Platform, *React, FastAPI, Gemini, Python, Pydantic, RAG* **Jan 2026**

- Designed an end-to-end claim intelligence workflow by integrating damage **photo analysis**, uploaded policy data, and itemized claim estimation logic, producing structured severity outputs and policy-adjusted repair cost projections
- Engineered **FastAPI** backend services by async image analysis endpoints and document extraction for receipts & insurance policies using JSON prompting & **Pydantic** validation, enabling consistent results across the **AI pipeline**
- Developed policy-aware claim calculation and legal guidance by **Xactimate-style** pricing, deductible and coverage logic tied to user's policies, and statute-grounded **RAG Q&A**, improving estimate interpretability and claim readiness

Ability Bridge — Computer Vision Accessibility Interface, *Python, OpenCV, MediaPipe, PyAutoGUI* **Sep 2025**

- Built a webcam-only assistive input system using **MediaPipe** FaceMesh (468 landmarks/frame, ~30 FPS $\approx$ 14,000 landmarks/sec) for hands-free cursor control and Morse-code typing.
- Stabilized real-time control using **solvePnP** head-pose tracking plus **EMA smoothing**, hysteresis thresholds, and debouncing to reduce jitter/false triggers; implemented eyebrow-click and long-blink commands via **PyAutoGUI**.

Experiences & Involvements

SASE Intern - UF Society of Asian Scientists & Engineers (SASE)

Aug 2025 - Dec 2025

- Coordinated **150+** attendee professional events by owning logistics, scheduling, and cross-functional stakeholder communication; supported speaker/vendor coordination and ensured smooth day-of operations.

Event Committee Member - UF Chinese American Student Association (CASA)

Aug 2025 - Oct 2025

- Spearheaded **300+** attendee cultural festival, overseeing project planning, stakeholder coordination & onsite execution.

Instructor - Logic Lab

June 2022 – July 2022

- Delivered math/chess instruction to **30+** students; co-developed lesson plans with a 5-person team..